

Redline Operations Manual

The self-serve guide to the production floor, QC, dispatch and admin screens that run Legend of Toys.

- OPERATOR
- SUPERVISOR
- DISPATCH
- ADMIN

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PART 1

Getting Started

What Redline is, how to sign in, find your way around, and what your role can see.

Welcome to Redline

- OPERATOR
- SUPERVISOR
- DISPATCH
- ADMIN

Redline is the production-floor cockpit for Legend of Toys. It is where the work that happens on the line (assembling, quality-checking, packing and dispatching every car and remote) turns into live numbers you can watch, plan against and act on. This manual is your self-serve guide to every screen in it.

What Redline is (and isn't)

Legend of Toys runs on three connected tools. They share one database, so an action in one shows up instantly in the others:

Scanner	The handheld app on the floor. This is where the physical work is recorded : an operator scans a unit at each station (inward, QC, packing, dispatch). Redline does not replace the scanner; it <i>reflects</i> it.
Garage	The store & inventory side: goods receiving, stock, the bill of materials, production-run requests and issuing parts to the floor.
Redline	This tool. The production & dispatch dashboard. It plans runs, designs and sets up lines, shows live output and quality, and drives the dispatch pipeline through to shipment.

① KEY IDEA

Most Redline screens are **read-only windows** onto what the floor is doing. The numbers come from scans. If a number looks wrong, the question is almost always “what was scanned?”, not “what did Redline do?”. A handful of screens (Line Setup, Returns, Corrections, Shipments, Repack Runs) *do* let you take action; each is called out clearly in its chapter.

The journey of a unit

Almost everything in Redline tracks a unit moving along this path. Knowing it makes every screen easier to read:

- 1 **Inwarded (INW):** a finished car or remote comes off assembly and is scanned in against a production run.
- 2 **QC Pass / QC Fail:** it is checked. Passes move on; fails go to the repair pool. (*See the QC chapter.*)

- 3 **Packed (PKG):** the car and its paired remote get their final sticker for a channel (Retail or Ecom).
- 4 **Dispatch:** the unit is allocated to a sales channel, packed into a shipment box, and shipped out.
(See the *Dispatch* chapters.)

Returns, repairs and channel-swap repacks are detours off this main path, covered in their own chapters.

How to use this manual

Each chapter covers one Redline screen and follows the same shape: **what it is for**, **when you'd open it**, **how to read every part of it**, and **what to do** with what you see. You do not need to read cover to cover; jump to the screen you're on using the Contents page or the bookmarks panel in your PDF reader.

Role tags

Every chapter is tagged with the people who typically use that screen. If you only do one job, you can skim past chapters that aren't tagged for you.



The next chapter, **Roles & What You Can See**, explains exactly what each tag means and which screens it unlocks.

Callout boxes

Coloured boxes flag things worth slowing down for:

✓ TIP

A faster or smarter way to do something.

✖ ON THE FLOOR

Practical guidance for using a screen live, on the production-floor monitor or a phone.

⚠ WATCH OUT

A common mistake or a number that is easy to misread.

⊖ IMPORTANT

Something that affects real stock, shipments or money: get it right.

Signing In & Getting Around

- OPERATOR
- SUPERVISOR
- DISPATCH
- ADMIN

Getting into Redline takes one click with your work Google account. Once inside, everything is reachable from a single sidebar grouped by the kind of work it does. This chapter shows you how to sign in and find any screen in seconds.

Signing in

Open `redline.legendoftoys.com` and choose **Sign in with Google**. Use your `@legendoftoys.com` account; only company accounts are allowed in, so a personal Gmail will be refused.

⚠ CAN'T GET IN?

If sign-in bounces you back to the login screen, you are almost certainly signed into a personal Google account in that browser. Sign out of Google entirely, or open Redline in a fresh browser profile, then sign in with your Legend of Toys account.

The layout

Sidebar (left)	Your main menu. It is collapsed to icons by default; click the toggle to expand it to full labels. Items are organised into the groups described below. Two items show a live count badge: Alerts (red) and Returns (orange).
Top bar	Shows where you are, and a "last refreshed" time so you know how fresh the numbers are. Many screens refresh themselves automatically.
Your account	Bottom of the sidebar: your name, your role, and Sign out .

The menu groups

The sidebar mirrors the parts of this manual:

GROUP	WHAT LIVES HERE
PRODUCTION	Dashboard, Planner, Line Design, Line Setup, Lines, Manpower, Hourly, QC, Audit, Deviations.

GROUP	WHAT LIVES HERE
ACTIVITY	Alerts, Returns, Scans, Corrections.
DISPATCH	Overview, Lines, Pipeline, Shipments, Delivery Challans, Channel Master, Repack Runs.
REPAIR	Customer Repairs, Repair Queue.
REPORTING	Reporting.
ADMIN	UPC Generator, Operators, Print.

You will only see the groups and items your role is allowed to open; see the next chapter.

Handy shortcuts

- **Quick search:** on any screen with a search box, press **/** to jump straight into it without reaching for the mouse.
- **Refresh:** most screens update on their own every 30–60 seconds. Where there is a manual  **Refresh** button, use it after you know something just changed on the floor.

✧ ON THE FLOOR

Redline is built to live on the big production-floor monitor as well as a laptop. The screens you'll leave running there are **Dashboard**, **Lines** and **Hourly**; they auto-refresh, so they stay current without anyone touching them. On a phone, stick to read-only screens; the action screens are easier on a larger display.

Roles & What You Can See

OPERATOR

SUPERVISOR

DISPATCH

ADMIN

Redline shows each person the screens that match their job. This keeps the floor team focused on what they need and keeps configuration screens in the hands of managers. This chapter explains the four roles used throughout this manual and which screens each one reaches.

The four roles

OPERATOR	Operator / Floor Team. Assembly, QC and packing staff. Mostly works from the scanner; in Redline, reads live screens to see how the day is going.
SUPERVISOR	Supervisor / Line Lead. Sets up lines each morning, assigns people, watches output and quality, and handles returns and repairs.
DISPATCH	Dispatch Team. Builds shipments, packs boxes, manages channels and challans, and runs channel-swap repacks.
ADMIN	Manager / Admin. Full access: everything above, plus line design, operator management, UPC generation, printing, reporting and configuration.

① HOW ACCESS ACTUALLY WORKS

Your access is decided by the **role on your account**, set by an admin. The four tags in this manual are plain-language groupings of those roles; they describe *who a screen is for*, not a literal setting. If you should be able to open a screen but can't, your account role needs changing; ask an admin rather than looking for a setting on the screen itself.

Who sees what

A quick map of the screens in this manual against the roles that typically use them. A dot means “this is part of your job”.

SCREEN	OP	SUP	DIS	ADM
Dashboard, Lines, Hourly	.	.	.	.
QC	.	.		.

SCREEN	OP	SUP	DIS	ADM
Planner, Line Setup, Manpower		•		•
Audit, Deviations		•		•
Alerts, Scans	•	•	•	•
Returns		•	•	•
Corrections		•		•
Dispatch Overview, Lines, Pipeline		•	•	•
Shipments, Challans, Repack Runs		•	•	•
Channel Master				•
Customer Repairs, Repair Queue		•		•
Reporting, UPC, Operators, Print				•

✓ **TIP**

This table is a guide, not a rulebook; some people wear more than one hat (a supervisor who also dispatches, for example) and will have a role that combines access. The dots show the *primary* owner of each screen so you know who to ask about it.

PART 2

Production

Planning runs, designing and setting up lines, watching output, and quality control.

New Run / Request

/new-run

OPERATOR

SUPERVISOR

ADMIN

New Run / Request is the single place production asks the store for anything. Tabs for the four run types, **Fresh**, **Outsourced**, **Repair** and **Repack**, plus an **Ad Hoc Parts** request for one-off part needs. Whatever you request lands in the store's Issue Queue, so there is one consistent "what does the store owe right now" list. (The Ad Hoc Parts tab shows only for production roles; store roles do not raise ad-hoc requests.)

WHAT IT'S FOR	READ OR ACT?	WHO	WHERE IT GOES
Requesting any run: Fresh, Outsourced, Repair, Repack.	Action.	OPERATOR SUPERVISOR ADMIN with run-request access.	The store's Issue Queue (Garage).

Production owns every run request. The store never issues stock on its own; it only fills a request that lands in its queue. Each tab below explains how the materials move, step by step, and which system handles each step.

Fresh

A normal in-house build. Pick the product, line, shift and the variant quantities (E-Comm and Retail). The run is created and submitted to the store; the store issues the full pick list and the floor builds it on the line. Nothing about fresh runs changed.

Outsourced

Raw materials go to a vendor who builds the units and returns them. This is now a **two-phase** run, so only the parts the vendor needs go out, and the packaging stays in until the units come back.

BUILD (outside)

1. Request Outsourced Run (pick vendor) -> store Issue Queue

2. Store issues the BUILD parts -> Car / Remote / Fastener only

3. Hand to vendor, then Send to Vendor (Garage)-> run goes In Progress

VENDOR builds. Units come back (in instalments is fine).

RECEIVE (store)

4. Receive built units (count) into the pool -> Garage run detail

FINISH (in-house, when a line is free)

5. Issue Finish Parts -> packaging / accessories / battery / sticker

6. Sticker each unit at Ext Inwarding -> then QC -> Packaging -> PKG Out

① ONLY CAR, REMOTE AND FASTENER GO TO THE VENDOR

The packaging, accessories, battery and sticker are issued later, in-house, when the built units return. You receive the returns as a simple count (instalments are fine), then issue the finish parts and sticker each unit at Ext Inwarding.

Repair

An open-ended recovery run for the broken / returned pile. You can start it with **no target**: just open the run and the floor pulls units in as they go. Add target product lines only if you already know what you are recovering.

1. Request Repair Run (no target needed)
2. Floor: at Repair Start, sticker each repairable unit and scan it in
(scrap goes straight to the scrap pile, no scan)
3. Repair it; when a part is short, raise an Ad-Hoc Request (Garage)
linked to this repair run -> store Issue Queue
4. QC -> Packaging -> PKG Out (QC fail -> Workshop -> QC, as normal)

① AD-HOC PARTS ARE TRACKED PER RUN

Parts requested for a repair run are linked to that run (and product) in Garage's Ad Hoc Request screen. Over time this builds a picture of which parts repairs really need, so we can buy extra and eventually structure the run.

Repack

Swap a unit's channel packaging when you need stock in one channel but it is packed for the other. Enter the product, the **from** and **to** channel, and the quantity.

1. Request Repack Run (product, from -> to channel, qty)
-> raises TWO requests:
 - (a) store: issue the TO-channel packaging (box + tray) -> Issue Queue
 - (b) dispatch: release the units -> "to release" list
2. Dispatch: at Repack Release, scan each box to hand it over (sign-off)
3. Repack In: scan the old box label (only released units)
4. Repack Out: new flipped label (channel set from the request) -> PKG Out -> dispatch

⚠ UNITS MUST BE RELEASED BY DISPATCH FIRST

This is the only place material moves backwards in the factory, so it is tightly controlled. A unit sitting with dispatch (Dispatch In or Allocated) cannot be repacked until a dispatch operator releases it at the new **Repack Release** station. Repack In hard-rejects anything that was not released. Units still on the production side (just out of PKG Out) do not need a release.

Ad Hoc Parts

A one-off parts request to the store, outside any run, for when the floor runs short. It is not a run (no units are produced); it just pulls parts. Build the request two ways:

- **BOM-based.** Pick a product and a part category; every part in that category is added, then you set quantities.
- **Manual.** Type each part code; the name fills in from the parts master.

Set line, shift and date, and optionally **link a repair run** so the parts are attributed to it (the repair feedback record). The request lands in the store’s Issue Queue like everything else.

① PRODUCTION ONLY

Raising ad hoc parts is a production job, so the Ad Hoc Parts tab appears only for production roles. It moved here from Garage so production asks the store for everything, runs and parts, in one place.

Recent Runs

Below the request tabs is **Recent Runs**, where you track and act on the runs you have requested, grouped by stage:

GROUP	WHAT’S HERE, AND WHAT YOU CAN DO
Requested	Created but not yet issued by the store. You can Cancel a run before it is issued.
Issued	In-house runs the store has issued. Confirm Receipt (accept the issued quantities; a shortfall automatically raises a short-supply request for the store), and Mark Complete when the run is done.
Upcoming	Outsourced runs sent to the vendor, waiting for the built units. When the store has inwards units, Request Finish unlocks and pulls the finish parts into the store’s Issue Queue. The button stays greyed until units are inwards.

Open ad-hoc parts requests are listed here too, with a Cancel action. Click any run to open its detail: vendor info, work orders, the pick list and the receipt status.

① THE STORE FULFILS FROM GARAGE

Issuing parts, rejecting a run, sending an outsourced run to the vendor and receiving the built units are the store’s steps, in Garage’s Issue Queue. You request and track here; the store fulfils there.

Dashboard

/exec

OPERATOR

SUPERVISOR

DISPATCH

ADMIN

The Dashboard is Redline’s home screen and the factory’s headline view for the day: how much has been inwards, passed QC, packed and dispatched, per line, against target. It refreshes itself every 30 seconds, so it’s built to live on the production-floor monitor all shift.

WHAT IT’S FOR	READ OR ACT?	COVERS	UPDATES
The day’s production at a glance.	Read-only.	The selected date.	Every 30s.

Today at a Glance

The top row of cards is the pulse of the day. The ones to watch:

CARD	MEANING
QC Pass / Inwards, Today	Units checked-and-passed, and units brought off the line, today.
QC Fail	Failed QC today (turns red when above zero). Sub-line splits cars vs remotes (e.g. 4C · 1R).
Pass Rate	First-pass yield. Green 95%+, yellow 85%+, red below.
Dispatched, Today	Units shipped (green); sub-line splits retail vs ecom.
PKG Out	Units packed and sitting ready-to-dispatch (at RTD).
Repair Queue	Units in the workshop (turns yellow above 50).
Open Findings	Outstanding audit findings (only shows when above zero): a prompt to visit Audit.

WTD / MTD Pass and Open Runs round out the row with week- and month-to-date passes and the count of live runs.

Plan vs Actual, Today

One card per production run, showing how far it has progressed against its target: a big % **complete** (green 90%+, yellow 60%+, orange above 0, red at 0), a progress bar, and a stats row: **Target, QC**

(units passed), **RTR** (ready-to-retail) and **RTE** (ready-to-ecom). The footer reads either -N short in red or ✓ On track in green.

Hourly Output

A per-line, hour-by-hour grid of dispatch achievement against target. Each line (L1 yellow, L2 blue, L3 green) shows its hourly target, today's total, and a pace label: ▲ ahead (green), ▼ behind (red), or "on pace". Each hour cell is colour-filled against its target: green met it, amber was close, red missed, and a corner dot marks an hour that beat target. Hours past 6 PM are marked OT (overtime).

Production Runs

A table of runs split into **Active Today** (green dot: today's Issued / In Progress runs) and **Upcoming** (blue dot: Submitted runs). Columns: Run, Product, Line, Date, Target, Retail, Ecom, Status.

▲ ABOUT THE DATE PRESETS

The **Today / This Week / This Month** chips change the date, but the cards, Plan-vs-Actual and runs all read the single selected *end* date; they don't add a whole week into one total. Use the presets to jump *to* a date, not to sum a range. For range totals, use **Reporting**.

✖ ON THE FLOOR

This is the screen to leave running on the wall. "Pace" and "behind/ahead" only appear once the shift is underway and a target is set, so an empty pace early in the morning is normal.

Planner

/planner

SUPERVISOR

ADMIN

The Planner turns the month’s dispatch demand into scheduled production runs. You upload the demand plan, the Planner shows what’s needed by which date and how much current stock covers it, and you build and create the runs to fill the gaps, without leaving the page.

WHAT IT’S FOR	READ OR ACT?	WHO	STARTS WITH
Scheduling production to meet dispatch demand.	Action: creates real runs.	SUPERVISOR ADMIN	A demand-plan CSV upload.

Step 1: Upload the demand plan

Use **↑ Upload Plan CSV** at the top right. The file must be the **Projection V2 tab** of the Dispatch Plan Google Sheet, exported via *File → Download → CSV*.

USE THE RIGHT TAB

Exporting the wrong sheet tab is the most common error; the upload will reject it with a message about a missing SKU, Channel or Mapping column. Rows with a dispatch date in the past are silently dropped, and an all-zero or unrecognised-mapping file shows “No demand rows found”. Always export the **Projection V2** tab.

The three tabs

Dispatch Timeline

What’s needed, by date. Each dispatch date is an expandable card showing total units, product count, and **⚠ N gap** in red where stock falls short. A toggle switches between **Cascaded Waterfall** (stock carries forward across dates per product) and **Normal** (each date against current stock). Expanding a product shows its variants with columns Need, RTD, Allocated, Balance/Available, Gap and a status pill.

Recommended Runs

The working tab. On the **left**, dispatch needs by date, each with a suggested production date (or **⚠ Too late to produce in time**) and a **+ Schedule** button per product. On the **right**, your **Production Schedule**: the “carts” you’re building.

- 1 Click **+ Schedule** on a product. A panel opens with editable Qty Ecomm / Qty Retail per variant (each showing how much gap remains), a production date, and an L1/L2/L3 line.

- 2 **Add to Schedule.** The line drops into a cart on the right. Build up as many lines and days as you need (+ **New Day** starts a fresh cart).
- 3 **Adjust in the cart.** Edit quantities inline;  repeats a line to another day,  removes it. A yellow  Stock short warning flags any line whose parts are insufficient.
- 4 **Create All for the date.** One button creates a real production run for every line in that cart with a quantity above zero. Each line reports  + run number, or an error.

Batch Sizes

Set the default run size per product (with the most common past size shown alongside). This drives how many runs the Planner suggests to cover a gap. Edit inline and Save.

Reading the status pills

PILL	MEANING
Covered	No gap: stock meets the need.
Plan Run	A gap exists; schedule a run with comfortable lead time.
Urgent	Tight on time: schedule today.
Critical / Too Late	Today-only, or already past the point of producing in time.

WATCH OUT

A product can't be scheduled on the same line and date twice; the Planner blocks it. Creating runs sends one request per line, so a large cart takes a few seconds; if a duplicate/open-run already exists you'll be asked whether to create another anyway or skip it.

Line Design

/line-design

ADMIN

Line Design defines the physical layout of each product’s assembly line: how many work stations sit in each department and how many people each holds. It’s the template that Line Setup staffs every morning. Designs are versioned, so layout changes are tracked over time.

WHAT IT’S FOR	READ OR ACT?	WHO	FEEDS
Designing the station layout per product.	Action: create/edit/version designs.	ADMIN / floor supervisors.	Line Setup’s staffing.

The screen

On the **left**, the list of products that have a design, each with its active version (e.g.

v3

) and effective date. On the **right**, the editor for the selected product: a summary bar of worker totals per department, then the four department sections each holding station cards.

Departments are always **Prep → Assembly → QC → Packaging**, and stations are auto-coded

PR-01

,

AS-02

,

QC-01

,

PK-03

 by department and position.

Working with station cards

Capacity	Click the person icon to toggle a station between one worker (👤) and two (👤👤).
Car / Remote	Prep and Assembly stations can be tagged CAR (blue) or REM (purple) to mark which stream they handle. Click to tag, click again to clear.
Reorder / remove	Drag a card to reorder within its department; ✕ removes it; the “+ Add station” tile adds one.

Creating and versioning

- 1

+ **NEW** creates the first design for a product. Pick the product, an effective-from date, optionally copy stations from another product’s design, and add notes.
- 2

Edit, then SAVE CHANGES. The save button stays greyed out until you actually change something; nothing is saved until you click it.

3

NEW VERSION when the layout genuinely changes. It carries the current stations forward as a starting point and requires a later effective date and a change note. The old version is kept in **Version History**, read-only.

① **ONE DESIGN PER PRODUCT**

A product can have only one line design. Once created it disappears from the New-design list; change it with **SAVE CHANGES** for small edits, or **NEW VERSION** when you want to preserve the old layout in history. To start a similar product, use the “copy stations from” option.

Line Setup

/line-setup SUPERVISOR ADMIN

Line Setup is the morning staffing screen. For a chosen date it takes the operators who have clocked in and lets you place them on the physical stations of each line, guided by who's worked there before. When you're done, print a roster for the floor.

WHAT IT'S FOR	READ OR ACT?	WHO	NEEDS FIRST
Assigning clocked-in staff to stations.	Action: assign / unassign / print.	SUPERVISOR ADMIN	Attendance done; a line design.

The layout

Three columns (L1, L2, L3) side by side. Each shows the run scheduled on that line (or NO RUN), department chips reading staffed/total (green when full), and the station cards from that line's design with one slot per worker.

Assigning people

- + Assign on an empty slot.** A picker opens with a search box and a suggestion list. Suggestions show frequency dots (● worked here before, ●● 3+ times, ●●● 6+ times), so you can place experienced hands first.
- Pick a name.** They're assigned immediately, no confirmation. The × on a filled slot unassigns them, also immediately.
- Suggest All** auto-fills every open slot in a department with the best-matched available people in one go.
- Print Roster** opens a printable sheet for that line (enabled once at least one person is assigned).

▲ **CLOCK IN FIRST**

Only **clocked-in** operators appear as suggestions. If nobody is clocked in, you'll see "No clocked-in operators available". Run attendance in **Manpower** first. If a line has a run but no line design, you'll be prompted to create one in **Line Design**.

① NO UNDO PROMPT

Assign and unassign take effect instantly with no confirmation; a mis-click is fixed by simply re-assigning. An operator can only be on one line per day; placing them somewhere new moves them.

Lines (Live View)

/lines

OPERATOR

SUPERVISOR

DISPATCH

ADMIN

Lines is the live floor monitor: each production line’s real-time output today, the people on it, and how fast each station is moving. Like the Dashboard, it’s built to run on the wall and refreshes every 30 seconds.

WHAT IT’S FOR	READ OR ACT?	COVERS	UPDATES
Watching each line’s live throughput.	Read-only.	Today only, lines L1–L3.	Every 30s.

Line Performance

One card per line, each showing the product on it, a big colour-coded % **complete** against the run target, and a five-stat grid:

STAT	MEANING
INW	Units inwarded (sub-line shows remotes).
QC Pass (green)	Passed QC; car/remote split below.
QC Fail (red)	Failed QC; car/remote split below.
PKG (yellow)	Units packaged.
Dispatched (green)	Retail + ecom shipped (sub-line splits R · E).

The footer adds operator count, downtime minutes, percent done, target, first-pass yield (FPY) and the first-scan time. The big % is green at 90%+, yellow at 60%+, orange at 30%+, red below.

Operator Output

A per-operator table: INW cars/remotes, QC pass cars/remotes, QC fails, and pass rate (green 95%+, yellow 85%+, red below). This is how you see who is moving and who may be struggling.

Station Pace (Takt)

Average time per unit at each station (INW, QC, PKG, and the shared PKG_OUT) across the lines. **Faster is greener**: green up to 5 min, yellow up to 10, red beyond. A slow, red station is your bottleneck.

⚠ TAKT REFRESHES ONCE

The line stats and operator output refresh every 30 seconds, but Station Pace is measured only when the page *opens*, so it can look stale next to the live numbers. Reload the page to refresh the pace figures.

Manpower

/manpower

SUPERVISOR

ADMIN

Manpower is the people screen: who is on the floor right now, daily attendance, the day’s roster, and a running log of performance points. It’s organised into four tabs: one live view and three you work in.

WHAT IT’S FOR	READ OR ACT?	WHO	UPDATES
Staffing the floor and tracking attendance.	Live View reads; the rest act.	SUPERVISOR ADMIN	Live View every 60s.

Live View

Today’s headcount across L1, L2, L3, Dispatch, Store, Others and Unassigned, then operator cards under each line. It shows only people with an **open shift** (clocked in, not yet out); someone who has clocked out drops off. “On floor” (from attendance) and “assigned” (from the roster) are different counts, shown side by side. Refreshes every 60 seconds.

Attendance

Clock-in/out records for a chosen date. Columns include Clock In, Clock Out, Duration and Device. A shift with no clock-out shows an amber Open badge and gets a **Close Shift** button.

⚠ CLOSE SHIFT SETS CLOCK-OUT TO NOW

Closing an open shift stamps the clock-out at the current time and can’t be undone here. Use it to tidy up someone who forgot to clock out; the duration is calculated from it.

Daily Roster

Assign clocked-in people to lines and stations for the day. Set target headcounts per line/station at the top (pre-filled from the active run’s line design), then staff from the **Available** pool: the people who are clocked in but not yet assigned.

- **Drag** a chip (or several selected chips) onto a station section, or use the + picker.
- **Auto-assign** fills the target slots in order from the available pool, then reports how many were placed and how many slots were left open.
- × removes someone from a line.

① ATTENDANCE COMES FIRST

"Available" only lists people who are *clocked in*. If someone isn't showing, they haven't clocked in yet. Sort attendance before rostering.

Performance

Pick an operator to see their points total (green if positive, red if negative) and event history. **+ Add Event** logs a point change: choose points (positive for good, negative for poor), a category (Attendance, Quality, Behaviour, Output, Other), a date and a reason. Every event is stamped with who recorded it.

Hourly

/hourly

OPERATOR

SUPERVISOR

DISPATCH

ADMIN

Hourly is a heatmap of output through the day: how many units cleared Inward, QC and Packaging in each hour, per line. Open it to see whether the floor is pacing steadily or stalling, and exactly which hour things slowed.

WHAT IT'S FOR	READ OR ACT?	COVERS	UPDATES
Watching throughput build hour by hour.	Read-only.	The selected date, lines L1–L3.	Every 30s.

How it's laid out

Three stacked panels (**Inward**, **QC Pass** and **Packaging**) each showing the day's grand total on the right. Inside each, one row per active line (L1 yellow, L2 blue, L3 green) with a strip of hour cells from 9 AM through 6 PM (later hours get a  for overtime). Each cell is a little bar that fills against that line's busiest hour, with the count printed inside.

⚠️ EMPTY CELLS AREN'T ZERO

Hours that haven't happened yet render faint and empty; that means "not reached", not "zero produced". Only past hours with no bar are genuine zeros.

✖️ ON THE FLOOR

Read it top to bottom: if Inward is flowing but QC's hourly bars are short, units are stacking up at the QC bench. A line that goes quiet for an hour mid-shift is the signal to check what happened: a parts shortage, a breakdown, or a break that ran long.

QC

/qc

OPERATOR

SUPERVISOR

ADMIN

The QC screen is your daily quality scoreboard. It does not pass or fail units (that happens on the scanner during the quality check), but it turns every one of those pass/fail scans into a clear picture of **how fast QC is moving**, **how much is passing first time**, and **what is going wrong and where**. It refreshes itself every 30 seconds, so you can leave it open and trust the numbers.

WHAT IT'S FOR	READ OR ACT?	COVERS	UPDATES
Spotting quality problems on the floor as they happen, today.	Read-only: a live view.	Today, all lines (L1–L3).	Automatically, every 30s.

① WHERE THE NUMBERS COME FROM

When a unit is inwarded it gets an **INW** scan; when it's checked it gets a **QC Pass** or **QC Fail** scan, and a failure records one or more **defect codes**. This screen simply adds those scans up for today. Nothing here changes a unit; to actually pass, fail or re-check a unit, use the scanner on the floor.

QC Cycle Time: INW to outcome

This is how long units are spending in quality control: the time from a unit being inwarded (INW) to getting its QC Pass or Fail. Shorter is better; long cycle times mean units are piling up at the QC bench.

The summary cards

CARD	WHAT IT TELLS YOU
Avg Cycle Time	The average across all lines, with the number of units it's based on. Colour-coded: green up to 30 min, yellow up to 60 min, red beyond. This is your headline number.
Median	The middle unit's time. If the median is much lower than the average, a few slow units are dragging the average up.
Avg, QC Pass	Average time for units that passed (green). Usually the fastest.
Avg, QC Fail	Average time for units that failed (red). Often longer, as failures take more inspection.
Fastest	The quickest unit through QC today.

CARD	WHAT IT TELLS YOU
Slowest (Normal)	The slowest unit that still counts as normal; genuine outliers are excluded (see below).

⚠ THE OUTLIER STRIP

If you see a red strip saying “*N outlier units excluded*”, that’s deliberate. A unit that sat overnight or was set aside for hours would wildly distort the averages, so the screen fences those off and tells you how many it removed and what the longest one was. The averages above the strip are the *true* working pace; the strip is just honesty about what was set aside.

Per-line breakdown

Below the summary, each active line (L1, L2, L3) gets its own card with the same figures. Use this to compare lines; if one line’s average is red while the others are green, that’s where to look first.

First Pass Yield by line

First Pass Yield (FPY) is the single most important quality number: **the percentage of units that passed QC the first time**, without needing repair. Each row is a line + product + date, with a coloured bar and a percentage.

COLOUR	FPY	WHAT IT MEANS
Green	95%+	Healthy. The line is building right the first time.
Yellow	85–94%	Watch it. Something is slipping; check the defects below.
Red	under 85%	Act. Too many units failing first time; find the cause now.

✓ TIP

A red FPY plus a high QC-Fail cycle time on the same line is a strong signal: that line is both failing more *and* taking longer to do it. Pair the two sections together when you diagnose.

Top Defects

For each line, the most common defects today, split into two columns:

- **Functional** (red total): the toy doesn’t work right: motor, electronics, movement.
- **Visual** (orange total): how it looks: scratches, paint, stickers, finish.

Each defect appears as a small card showing its **code**, a short **description**, its **category**, a **severity** (Critical, Major or Minor), and how many times it occurred. Cards are ordered worst-first: Critical before Major before Minor, then by count. A **FLAG** marker means the defect is flagged for operator training.

Defect Breakdown by product

The same defects, organised by **product** instead of line. Each product is a row showing its total defect count; **click a row to expand it**. Inside, defects are grouped first by component (Car vs Remote), then by severity, then listed individually with their code, description and count. A **TRAINING** tag again marks defects flagged for coaching.

✖ ON THE FLOOR

Use *Top Defects* when you're asking "which line needs help right now?" and *Defect Breakdown* when you're asking "which product is giving us trouble?". Same data, two lenses.

Repeat Failures

The last section is a table of individual units that failed with **more than one defect**: the units giving the most trouble. Each row shows the unit's UPC, product, total defect count (red), how many unique defect codes it hit, and the list of codes.

🚫 IMPORTANT

A unit appearing here with many defects is a candidate for a deeper look before it's re-worked; repeated, varied failures on one unit can point to a damaged part or a build that should be pulled rather than patched. Cross-check its UPC in the **Scans** screen to see its full history.

A typical QC check-in

- 1 **Glance at the Avg Cycle Time card.** Green and steady? QC is keeping pace. Red? Units are backing up; check which line in the per-line cards.
- 2 **Scan the FPY bars.** Any red or yellow line is your priority.
- 3 **For a problem line, read its Top Defects.** Functional or visual? Critical or minor? That tells you whether it's a parts/assembly issue or a finishing issue.
- 4 **If a product keeps recurring, expand it in Defect Breakdown** to see exactly which defects and components are driving it.
- 5 **Check Repeat Failures** for any single units that need pulling rather than re-working.

QC Audit

/audit

SUPERVISOR

ADMIN

QC Audit is where supervisors run walk-the-floor quality rounds, record what they find, and chase every finding through to fixed. It has two tabs: **Log** for running rounds today, and **Tracker** for managing open findings over time.

WHAT IT'S FOR	READ OR ACT?	WHO	SEVERITY
Running QC rounds and resolving findings.	Action: rounds and findings.	SUPERVISOR ADMIN	Critical · High · Medium · Low.

Log: running a round

A day summary strip (rounds today, critical/high open, total findings, confirmed) sits above the round cards. To run a round:

- 1

+ **Start New Round.** A round opens, marked Active (green).
- 2

+ **Add Finding** for each issue. Pick a **Line**, a **Category** (e.g. QC Pass: Damage, Assembly: Wrong/Missing Part, SOP Deviation), a **Severity**, a description, and the action required.
- 3

Close Round when the walk is done. Closed rounds still expand, and their findings can still be resolved and confirmed.

Tracker: closing the loop

Filter all findings by line, status, severity, category and date. Per-line scorecards show open counts and the oldest open item; a card border turns red when its oldest open finding is **overdue**. The findings table carries the action buttons.

STATUS	MEANING & NEXT STEP
Open	Raised, not yet fixed. A supervisor marks it Resolved with a required note describing the fix.
Resolved	Fixed, awaiting sign-off. The round's auditor (or a super admin) clicks ✓ Confirm Fixed , or anyone can Reopen it.
Closed	Confirmed fixed, shown dimmed. Done, not gone.

A **Repeat Offenders** panel flags findings that have recurred in the last 30 days: the patterns worth fixing at the root.

SEPARATION OF DUTIES

The supervisor who *ran* a round cannot mark its own findings resolved; a different supervisor must. The round's auditor (or a super admin) then confirms the fix. Resolving always requires a written note of what was done; blank notes are rejected.

OVERDUE RULES

An open Critical or High finding is overdue after 24 hours; Medium or Low after 48. Overdue ages show red and turn the scorecard border red: your prompt to escalate.

Line Flush

/line-flush

OPERATOR

SUPERVISOR

ADMIN

A line flush is how production returns leftover material to the store at the end of a run: the unused, damaged, QC-rejected or partly-assembled parts that did not go into a unit. You raise the flush here; the store receives, dispositions and verifies it in Garage.

WHAT IT'S FOR	READ OR ACT?	WHO	THEN
Returning leftover material to the store.	Action: raises a flush.	OPERATOR SUPERVISOR ADMIN with flush access.	Store verifies it in Garage.

Raising a flush

- 1

Click **New Flush**. Set the date, line, shift, and pick the flush type: against a **Production Run** or **Standalone**.
- 2

Add the parts being returned. For a run flush, use **Load From Run** to pull the run's pick list, or add parts by code. Each part can be split across return types (Unused, Damaged, QC Rejected, Partial Assembly, Wrong Issue) with a quantity per split.
- 3

Submit Flush. It is sent to the store with a flush id, marked Pending Verification.

Tracking flushes

The list shows every flush with its status: **Pending Verification** (waiting for the store), **Verified** (the store counted and accepted it) or **Disputed**. Open one to see the parts you returned and, once verified, the store's counted quantities, any variance, and the dispositions they applied.

① VERIFICATION HAPPENS IN GARAGE

Raising a flush is production's job and lives here. Counting the returned material, setting dispositions (back to stock, quarantine, scrap, rework) and accepting it into inventory is the store's job, on Garage's **Flush Verify** screen. The store's physical count is the figure that actually moves stock, so the detail you see here is read-only.

Process Deviations

/process-deviations

SUPERVISOR

ADMIN

Process Deviations is the full record of process changes, temporary or permanent, and their supervisor review. It is where someone documents a change to how a step is done, routes it for the right level of sign-off, and acts on it through its whole life: propose, approve or reject, acknowledge, and close. This is purely a production tool; the store has no process deviations.

WHAT IT'S FOR	READ OR ACT?	WHO	TIERS
Documenting & approving process changes.	Full management: propose, approve, close.	OPERATOR proposes; SUPERVISOR ADMIN approve.	L1 / L2 / L3 by severity.

The status tabs

Tabs across the top filter the list: **Pending Approval**, **Active on Floor**, **Retro Sign-off**, **Rejected**, **Closed**, and **All**. The **Active on Floor** tab is laid out as cards grouped by line, so anyone can see at a glance what is in force on each line right now: each card shows the deviation number, severity, title, station or type, and when it expires or that it is open-ended. **Operators must follow active deviations**. Every other tab is a table you can click into.

Working a deviation

Click any row to open its full detail: the description, reason and rollback plan, line/run/product/station, the approval window, and a complete history of every action taken on it. From here, depending on your permission and the deviation's current tier, you can:

- **Approve or Reject** (reject needs a reason) when it is pending at your tier.
- **Escalate** a pending deviation to the next tier.
- **Acknowledge** an approved, rejected or closed deviation, so there is a record you have seen it.
- **Confirm or reject a retroactive sign-off** for a reactive deviation applied while you were out.
- **Close** an approved deviation (reason required) once it no longer applies.

Proposing a deviation

+ **Propose Deviation** opens a form: pick a **type** (Material Substitution, Step Skip, Quality Criteria Relaxed, Process Workaround...), a **severity**, the line and station, a title and description, an optional reason and rollback plan, and an optional "effective from / until" window. The selected severity shows its approval rule:

SEVERITY	WHO APPROVES
Low	L1 supervisor.
Medium	L1 + an L2 second pair of eyes.
High	L3 admin only.
Critical	L3 admin only; reactive auto-approval blocked.

Tick ⚡ **Reactive** if the change is already in effect and you are logging it after the fact. Low and medium reactive deviations are created as approved immediately but flagged for a supervisor’s retro sign-off; High and Critical still wait for L3 sign-off regardless.

① THE FULL RECORD LIVES HERE

This screen is the complete deviations queue and history, moved into Redline so production owns it end to end. The permissions that decide who can approve at each tier are still set in Garage’s Users screen (the company-wide role matrix); this screen simply reads them.

PART 3

Activity

Live alerts, returns intake, the raw scan log, and fixing mistaken scans.

Alerts

/alerts

OPERATOR

SUPERVISOR

DISPATCH

ADMIN

Alerts is the floor’s exception list: scans that were rejected or flagged by the rules, such as an over-target pack, a wrong-station scan, or a failed check. Open it to review what went wrong over a date range and mark each one as handled.

WHAT IT’S FOR	READ OR ACT?	WHO	BADGE
Reviewing and clearing scan violations.	Reads; one action: acknowledge.	SUPERVISOR ADMIN (+ all can view)	Red count in the sidebar.

Reading it

Four overview cards head the screen: **Total**, **Unacknowledged** (red when above zero), **Worst Line** and **Worst Station**. The table below lists each violation with its time, line, station (colour-coded), operator, the unit’s UPC, and the violation message in red. The status badge reads

● open

 (red, needs attention) or

✓ Ack

 (green, handled); acknowledged rows are dimmed.

Filter by status (All / Unacknowledged / Acknowledged), by line, and by station, and set the From/To dates (the **Today** chip resets to today).

Acknowledging a violation

- 1 Click **Ack** on an open row. A box shows the violation detail and an optional note field.
- 2 Add a note (optional) and **Confirm Ack**. The row turns green and dims; it’s marked as seen and dealt with.

① ACKNOWLEDGING ISN’T FIXING

“Ack” records that a supervisor has seen and handled the alert; it doesn’t change the scan or the unit. The aim each day is to drive **Unacknowledged** to zero, having actually dealt with the cause on the floor.

Returns

/returns

SUPERVISOR

DISPATCH

ADMIN

Returns is a floor view of the return piles — **UDR, CXR, BRV, Loss and Scrap** — grouped by variant, with KPI totals. From here Production can **request a repair run**. Dispositioning and issuing happen in Garage.

WHAT IT'S FOR	READ OR ACT?	WHO	UPDATES
Watching piles + requesting repair runs.	Mostly read; request a run.	SUPERVISOR DISPATCH ADMIN	Every 60s.

The piles

UDR	(green) Undamaged — re-dispatch as-is via PKG OUT, no repair.
CXR / BRV	(yellow / blue) Customer and bulk/vendor returns — go to a repair run.
Loss	(red) Switcheroo / wrong / not received — written off.
Scrap	(grey) Units scrapped on the repair line.

Each pile is grouped by product / model / colour with a count (and oldest age for UDR/CXR/BRV). The KPI cards sum each pile.

Request a repair run

Request Repair Run opens a picker of the CXR/BRV buckets and a line. It creates an empty

REP-NNN

 run with those target lines; the **Store** then physically issues (scans out) the units against it on Garage → Issue Repair.

① PAIRING IS RE-MADE AT QC PASS

Returned cars and remotes move through repair independently; a fresh car↔remote pair is created at the QC PASS station, like new production. Dispositioning and issuing are done in Garage.

Scans

/scans

OPERATOR

SUPERVISOR

DISPATCH

ADMIN

Scans is the master log of every barcode scan on the floor (inwarding, QC, workshop, packing, returns and more) with daily totals. Open it to trace a single unit’s full history, or to review a day, week or month of activity.

WHAT IT’S FOR	READ OR ACT?	COVERS	SHORTCUT
Looking up scan history and totals.	Read-only.	Any date or range; any UPC.	 focuses the UPC search.

Two ways to use it

- 1

Review a period. Pick a date or a Today / This Week / This Month preset. The summary cards count each activity for the day (INW and QC Pass also split cars vs remotes); click any card to filter the table to that activity. **Load More** pages through the list.
- 2

Trace a unit. Type a UPC into the search box. Typing **4 or more characters** searches the unit’s entire history across all dates; 1–3 characters just filters the rows already on screen.

Reading the table

Columns: Time, UPC, **Activity** (colour-coded badge), Line, Product, Operator, **Loop** (how many times the unit has cycled through), and Status: ☒ OK or ☐ Voided . Tick **Show Voided** to include cancelled scans (they show dimmed). Activity badges follow the floor’s colour key: INW blue, QC Pass green, QC Fail red, WKS purple, PKG violet, RTO/RTD orange-teal.

▲ TWO THINGS THAT TRIP PEOPLE UP

A **short** UPC search (1–3 characters) only filters the current day’s loaded rows; it won’t find scans on other days. Type the full UPC (4+ chars) for a true all-history lookup. Also, the summary cards always count the *start* date, even when a multi-day range is selected for the table.

Corrections

/corrections

SUPERVISOR

ADMIN

Corrections is the controlled way to fix a mistaken scan. Supervisors can **void** a wrong scan; managers can **amend** a scan’s line or notes. Every change is logged and kept; nothing is silently deleted.

WHAT IT’S FOR	READ OR ACT?	WHO	AUDIT
Fixing scans recorded in error.	Action: void & amend.	SUPERVISOR (void) ADMIN (amend)	Every change is logged.

The two tiers

ACTION	WHO	WHAT IT DOES
Tier 2: Void	Supervisor	Cancels a wrong scan made in the <i>current shift</i> . The scan stays in the log marked Voided. A reason is required.
Tier 3: Amend	Manager	Corrects a scan’s line or notes on any scan, leaving an immutable audit trail. A reason is required.

Doing a correction

Find the scan (pick the date, optionally filter by activity or search a UPC). Then use the action button on its row:

- 1 **Void.** Confirm the scan shown, type a required **reason**, and Confirm Void. The row becomes Voided.
- 2 **Amend.** Change the **Line** and/or **Notes**, type a required **reason** (the Confirm button stays disabled until you do), and Confirm Amend. It rejects “no changes to save” if nothing was actually changed.

▲ GOOD TO KNOW

You’ll only see the action buttons your permission allows: void needs supervisor rights, amend needs manager rights, and an already-voided row offers neither. Void is meant for the current shift; an older scan may be refused by the system even if the button shows. Amend here changes line and notes only.

PART 4

Scanner

The production and dispatch stations on the floor scanner:
how to get in, and what each station does.

The Floor Scanner



The floor scanner is the handheld PWA your team uses at every physical station. The whole production line and the whole dispatch line run through it. This chapter covers how the scanner is now organised: how you get into your department, sign in, and reach the station you need. The station chapters that follow cover each scan in detail.

WHERE	YOUR AREAS	WHO	GETS YOU IN
scanner.legendoftoys.com , on the floor tablets.	Production and Dispatch.	  	Department PIN, then your QR badge.

Departments, not a free-for-all

The scanner opens on a **Department Select** screen with four tiles: **Production**, **Store**, **Dispatch** and **Attendance**. Each tile leads only to its own stations. A Production device can no longer be switched to a Store or Dispatch station by mistake, and the reverse. This walling-off is deliberate: it removes the wrong-station mis-scans that used to cause real damage, like fresh cars being scanned through a returns station.

- 1 Tap your department.** Production or Dispatch. A 6-digit PIN keypad appears on screen (not the phone keyboard, so the PIN is never remembered by the device).
- 2 Enter the department PIN.** A wrong PIN just says “try again”; there is no lockout. The unlock then sticks until **1 AM**, even if the app is closed and reopened.
- 3 Sign in with your QR badge.** Scan your operator card (or tap **Type ID** and enter your employee ID). This runs once per device per shift and also expires at 1 AM.
- 4 Pick a category, then line and station.** Categories show stations in floor order. Choose your line (**L1–L5** for production, **D1/D2** for dispatch) and station, then tap **Launch Scanner**.

▲ **PRODUCTION NOW NEEDS A SIGN-IN TOO**

Every working station, including all Production stations, now asks for your operator QR badge once per device per shift, the same as Dispatch always has. Scan your card after setup. Only the device-free utilities (Lookup and Attendance) skip this.

On the scanner screen

Once launched, the top bar shows your **line and station** (and operator name) on the left and **Logout** on the right. The camera fills the middle; your recent scans run along the bottom. There is **no settings button and no way to switch station in-app**. To change line or station, tap **Logout**, which takes you back to Department Select, then come in again through the guided flow. That friction is intentional.

Shift is automatic

There are no Regular / OT buttons any more. The scanner reads the device clock and tags each scan's shift for you: Packaging counts Regular from 10:00 to 19:00, every other station from 09:00 to 18:00, and anything outside that window as OT. You never set it.

When a scan is rejected

A WRONG SCAN STOPS YOU WITH A FULL RED SCREEN AND A BUZZ

If a unit is at the wrong stage, belongs to a different run, or the code is the wrong kind, the scanner shows a **full red reject screen with a distinct buzz** and states the exact reason (for example "wrong run, unit belongs to REP-123"). Nothing is recorded. Read the reason, fix it, and rescan. Every station below has been tightened to reject the wrong scan rather than let it through quietly.

Production Stations

OPERATOR

SUPERVISOR

The Production department’s **Fresh Production** category walks a unit along the line in floor order: Assembly, QC Pass, QC Fail, Workshop, Packaging, then PKG Out. Each station accepts only a unit that is genuinely at the right stage; anything else is rejected with a red screen.

CATEGORY	STATIONS	WHO	YOU SCAN
Fresh Production · lines L1–L5.	Assembly → QC → Workshop → Packaging → PKG Out.	OPERATOR SUPERVISOR	The LOT UPC, or a box label at PKG Out.

Assembly (INW)

Brings a freshly built unit into the system. Scan the unit’s

LOT-XXXXXXX

 UPC; a green tick confirms it is inwards. Assembly checks the unit’s product against the line’s active run, and that check now applies to **remotes as well as cars**, so a remote from the wrong product is caught the same way a wrong car is. A code that is not a LOT UPC, or a unit already inwards, is rejected.

QC Pass and QC Fail

The quality check. Scan the unit’s UPC only. A unit can be checked only if it has been **inwards** (fresh) or **repaired** (back from a repair run); anything else is rejected. QC Pass marks it good; QC Fail then asks you to pick the defect codes. On a passing car-plus-remote, the pair is recorded here.

Workshop (WKS)

Where a failed unit is worked on. Scan the unit’s UPC. After workshop, a unit goes back through QC.

Packaging (PKG)

Boxes a finished unit. Scan the **car UPC first**, then its remote. Two rules are enforced:

- **QC Pass only.** Only a unit that has passed QC can be packed. A unit at any other stage is rejected.
- **The right pair.** The remote you scan must be *this car’s* actual paired remote, not just any remote.

⚠ RE-PACKING IS NOT DONE AT PACKAGING

Packaging is for fresh units only. To re-pack a unit that has already been boxed or dispatched (for example to swap its channel), use the **Repack** stations, not Packaging. Packaging will reject an already-dispatched unit.

PKG Out

The hand-off from production to dispatch. Scan the box label (`LOT-XXXXXXX-E/R`). PKG Out accepts exactly two things:

- a **fresh** unit just boxed at Packaging (awaiting dispatch), or
- a **UDR return** that the Store has issued back out for re-dispatch.

A UDR return keeps its original sealed box, so PKG Out re-dispatches it on that same label without re-boxing. Anything that is neither is rejected.

Repair, Outsourced & Repack

OPERATOR

SUPERVISOR

Three more Production categories handle units that are not fresh builds: **Repair Run** for units coming back for rework, **Outsourced Run** for finished units returned by a vendor, and **Repack Run** for swapping a packed unit’s channel. Each starts at its own first station and then rejoins the normal QC-to-PKG-Out line.

CATEGORIES	FIRST STATIONS	WHO	THEN
Repair Run · Outsourced Run · Repack Run.	Repair · Ext Inwarding · Repack In.	OPERATOR SUPERVISOR	Rejoin QC → Packaging → PKG Out.

Repair Run

A repair run (REP-NNN) is a batch of returned or pile units to be reworked. You select the run, then scan each unit in at the **Repair** station to start it. A returned unit issued from the Store’s Issue Repair pool is accepted only if it was issued to this exact run. A loose pile unit with no valid label gets a **fresh sticker** applied first, then scanned in. The Repair scan also marks the unit as having been repaired, so anyone downstream can tell a repaired unit from a fresh one by its history. From there the unit goes through Workshop and QC like any other; on a QC Pass the car and remote are re-paired exactly as in fresh production. Parts the repairer needs are requested off the floor as an Ad-Hoc Request linked to the run (see Garage).

Outsourced Run

An outsourced run (EXT-NNN) sends raw materials to a vendor who builds and returns finished units. New outsourced runs are **two-phase**: only car, remote and fastener go out; packaging and accessories are issued in-house when the units return. **Ext Inwarding is the finish step**: once the store has issued the finish parts, scan each returned unit in here to sticker it and bring it into the system against the run; it then flows through QC, Packaging and PKG Out like a fresh unit. Ext Inwarding checks that the scanned unit’s product matches the run, so a unit cannot be tagged to the wrong job. (Older outsourced runs created before this change keep their original one-step receive.)

Repack Run

A repack run swaps a packed unit from one channel to another. It is two stations:

- **Repack In.** Select the run, then scan the unit’s **box label** (LOT-XXXXXXXX-E/R). It accepts a box label only, and only a unit that is either just out of **PKG Out** (still on the production side) or one that dispatch has **released at Repack Release**. A dispatch-held unit (Dispatch In or Allocated) that was

not released is rejected, and a shipped unit or a bare car / remote is always rejected. Scanning it in releases any allocation and rolls it back so it can be re-boxed for the new channel.

- **Repack Out.** Scan the unit and its destination box. Only a unit actually scanned in at Repack In is accepted, so a fresh unit cannot slip through. The new channel is set automatically from the repack run's request, so you do not pick it by hand.

⚠ **DISPATCH UNITS MUST BE RELEASED FIRST**

If the units are with dispatch, a dispatch operator hands them over at **Repack Release** (Dispatch department) before Repack In will take them. Units fresh from PKG Out skip that step. The store also issues the new channel's packaging (box and tray) into the Issue Queue when the run is created.

① **SAME HANDLERS, DIFFERENT DOORWAYS**

Repaired and outsourced units flow through the *same* QC, Packaging and PKG Out stations as fresh units; the categories are just separate doorways into one line. So a repaired or vendor-built unit is dispatched exactly like a freshly built one once it passes QC.

Dispatch Stations

DISPATCH

SUPERVISOR

The Dispatch department’s **Dispatch Flow** category moves a finished unit out the door in floor order: Dispatch In, Allocate, Pack, then Dispatch Out. A separate **Restock** station pulls an unused unit back into stock. Dispatch stations bind to a dispatch line, D1 or D2.

CATEGORY	STATIONS	WHO	YOU SCAN
Dispatch Flow · lines D1/D2.	Dispatch In → Allocate → Pack → Dispatch Out.	DISPATCH SUPERVISOR	Box labels (LOT-...- E/R) and box refs.

Dispatch In (DTK)

Takes a packed unit into the dispatch area. Scan the box label (LOT-XXXXXXX-E/R). It accepts only a unit that has come through **PKG Out** (ready to dispatch); anything else is rejected.

Allocate (ALLOC)

Assigns a unit to a channel or shipment. Select the channel, then scan the box label. It accepts a unit at **Dispatch In** (first allocation) or one already **Allocated** (re-allocation is free), so you can move a unit between channels without a detour.

Pack (PACK)

Builds the physical shipment box. Select the shipment, then scan each unit’s batch label into it. The shipment dropdown shows **all open shipments**, not just today’s, so you can pack ahead for a future-dated shipment.

Dispatch Out (DOUT)

The final out-the-door scan. Scan the box label (BOX-XXXXX) or a batch label to ship it. After this the unit is shipped.

Repack Release

Hands a unit back to a **repack run** so it can be re-boxed for a different channel. When production raises a repack run, the units it needs show up as a release list. Select the run, then scan each unit’s **box label** (LOT-XXXXXXX-E/R) to release it. Releasing locks the unit out of the normal dispatch flow and lets it be

scanned at Repack In. Only units sitting at **Dispatch In** or **Allocated** need releasing here; a unit still on the production side does not.

▲ THIS IS THE CONTROLLED HAND-OVER

Nobody should walk into the dispatch area, grab a unit and start repacking it. Repack In will reject any dispatch-held unit that was not released here first. Scanning it out at Repack Release is the proper, signed hand-over, one box at a time.

Restock

Returns an **unused** unit to stock (for example, a unit pulled from a box that was over-packed). Pick a reason, then scan the unit's LOT UPC. Only unused units are accepted; this is not the path for a customer return.

▲ A COURIER-TOUCHED RETURN NEVER GOES THROUGH RESTOCK

If a unit has been out to a customer and come back, it is a return, not a restock. Send it through **Returns Intake** in the Store department. Restock is only for unused units that never left.

Attendance & Lookup

OPERATOR

SUPERVISOR

DISPATCH

Two scanner tools sit outside the working flow: **Attendance**, for clocking staff in and out, and **Lookup**, for checking any unit’s history. Neither moves production along and neither needs an operator sign-in.

TOOLS	YOU SCAN	WHO	RISK
Attendance Lookup	An employee QR card, or any unit code.	OPERATOR SUPERVISOR DISPATCH	Both are safe, no sign-in.

Attendance

Attendance is its own tile on the Department Select screen and opens straight into clock-in / clock-out, with **no PIN, no line and no operator gate**. Scan the employee’s QR card (or tap **Type ID** and enter it) to record the clock event. A 4-digit ID auto-submits; press Enter for a longer legacy ID. Back returns to Department Select.

Lookup

Lookup is the safe “what is this unit?” tool, available under the Utilities category in both Production and Dispatch. Scan any code, a LOT UPC, a box label, or a remote, and it shows that unit’s full history, box and packing information. It changes nothing, so use it freely to check a unit before acting on it elsewhere.

✕ WHY THESE TWO SKIP THE SIGN-IN

Every working station now asks for an operator badge, but Attendance and Lookup are device-free utilities that record no production work, so they open without one. That also means you can reach Lookup quickly to settle a “where is this unit?” question mid-shift.

PART 5

Dispatch

From open inventory through shipments, channels, challans and channel-swap repacks.

Dispatch Overview

/dispatch

DISPATCH

ADMIN

The Dispatch Overview is the dispatch-floor home screen: how many packed units sit at each stage, what’s committed to which channel awaiting dispatch, what’s gone out recently, and a lookup for any individual unit. It refreshes every 30 seconds.

WHAT IT’S FOR	READ OR ACT?	WHO	UPDATES
The dispatch day at a glance.	Read-only.	DISPATCH ADMIN	Every 30s (units table on demand).

Live Dispatch Status

Four cards trace the dispatch pipeline a unit moves through:

CARD	MEANING
PKG Out	Packed on the line, awaiting handover to dispatch.
With Dispatch	Handed over, to be allocated to a channel (yellow when above zero).
Allocated	Committed to a channel, awaiting ship (blue when above zero).
Shipped	Total shipped (green).

By channel

Allocated: Awaiting Dispatch shows how many committed units wait per channel. **Sent Out by Channel** shows shipped units over a chosen window (10 Days, This Week, This Month or Custom), with a

sold · non-sale · total

 summary; revenue channels are highlighted in green, cost/non-sale channels dimmed.

Units lookup

The **Units** table lists individual units (Batch Label, Product, Line, Channel, Shipment, Packed date and Status), filterable by status, channel and date. Status badges: PKG Out (neutral), With Dispatch, Allocated (blue), Shipped (green ✓).

⚠ THE TABLE IS CAPPED

The Units table shows up to **200 rows** and doesn't auto-refresh; use the filters and the  **Refresh** button. On a heavy day, narrow by status or channel to find what you need.

Dispatch Lines

Dispatch Lines is today’s dispatch productivity board: how many scans happened at each dispatch stage per line, which channels each line is working, and who’s doing the work. It refreshes every 30 seconds.

WHAT IT’S FOR	READ OR ACT?	COVERS	UPDATES
Live dispatch-line & operator output.	Read-only.	Today only.	Every 30s.

The four dispatch stages

Every figure on this screen breaks down into the four dispatch-floor scan stages. Knowing them makes the board readable:

STAGE	MEANING
Alloc	Unit allocated (committed) to a sales channel.
DTK	Dispatch take: pulled to be boxed.
Pack	Packed into a shipment box.
Dout	Dispatched out: shipped.

Reading it

Dispatch Line Performance gives one card per line: its active channels, a big total-scans number, the four-stage grid, operator count and the day’s first-to-last scan time. **Operator Output** is a table of each operator’s Alloc / DTK / Pack / Dout and total, sorted by who’s done the most.

① “TEST OPERATOR” IS HIDDEN

A non-production “Test Operator” account is deliberately left out of the operator table so it doesn’t skew the rankings.

Dispatch Pipeline

The Dispatch Pipeline is your single answer to the question “what do we have ready to ship, and where is it committed?”. It’s one table showing every unit that is built but not yet shipped, broken down by product, by whether it’s still in production or finished, and by which sales channel it’s been allocated to. It’s the planning view dispatch and supervisors use to decide what to pack next.

WHAT IT’S FOR	READ OR ACT?	COVERS	UPDATES
Seeing all open (un-shipped) inventory in one place.	Read-only: a planning view.	Every product with open units.	On open, and on Refresh.

“OPEN INVENTORY” MEANS

Units that exist but have **not yet shipped**: anything between coming off the line and going out the door. Once a unit ships, it leaves this table. So a shrinking pipeline through the day is normal and healthy; a growing one means you’re building faster than you’re shipping.

Reading the columns

Every product is a row, and the columns split its open units into stages of readiness. Reading left to right walks a unit from “still being built” to “committed to a channel”:

Product / Variant	The product name. Click the row to expand it into its variants (model + colour). The little arrow (► / ▼) shows whether it’s open.
With Production	Units still in the production stage, built and inwarded but not yet finished and ready for dispatch . These are coming, but you can’t ship them yet.
Unalloc (R)	Unallocated, Retail . Finished, packed for the <i>Retail</i> channel, but not yet committed to a specific dispatch channel. Shown in yellow: this is stock waiting for a decision.
Unalloc (E)	Unallocated, Ecom . The same, but packed for the <i>Ecom</i> channel. Also yellow.
Total Allocated	Units already committed to a channel and waiting to be packed into a shipment (green). The sum of all the channel columns to its right.

Channel columns

One column per active sales channel (Amazon, Flipkart, Cred, retail partners...), showing how many allocated units are committed to each. These are the live channels from the Channel Master.

✓ READING R VS E

"Retail" and "Ecom" here are the **packing format** a unit was finished in (the sticker it carries), not its final destination channel. A unit is first packed Retail or Ecom, then later *allocated* to a specific channel. That's why a unit moves from the yellow Unalloc columns into a green channel column as dispatch commits it.

The totals row

The bold row pinned at the top of the table sums every column across all products: your whole open pipeline at a glance, with a count of how many products it spans. Read it first to size up the day, then drill into individual products below.

Expanding a product

Click any product row to reveal its **variants**: each model and colour combination on its own indented line, with the same columns. This is how you answer questions like "we have 40 of this product unallocated, but is it the *red* one the channel wants, or the blue?". Click again to collapse.

Putting it to work

- 1 Start with the totals row.** How much is finished-but-unallocated (yellow) versus already committed (green)? A big yellow number is stock you could be allocating and shipping now.
- 2 Find products with high Unalloc.** These are your easiest wins: finished units only waiting to be pointed at a channel.
- 3 Check the channel columns against demand.** If a channel needs units and a product has them sitting unallocated in the matching format, that's the next thing to allocate and pack.
- 4 Use With Production to plan ahead.** A low ready figure but a high With Production figure means supply is on the way; don't over-promise from current stock.
- 5 Expand a product to confirm the variant** before committing; the right total of the wrong colour won't fill the order.

⚠ WATCH OUT

A dash (—) means zero: the column is empty for that product, not missing data. And remember this view is a *snapshot*: it reflects scans up to the moment you opened it or last hit Refresh. After a busy packing burst, refresh before you make a call.

① WHERE TO ACT

This screen tells you *what* to do but doesn't do it. Allocating units to a channel and packing them into boxes happens on the **Shipments** screen; the channel list itself is maintained in **Channel Master**. Both have their own chapters in this part.

Shipments

/dispatch-shipments

DISPATCH

ADMIN

Shipments is the main dispatch workspace, where outbound orders are built and closed out. You create a shipment for a channel, set what it should contain (the manifest), open boxes and pack units into them, then mark it shipped. This is the one Dispatch screen where you do real work, not just watch.

WHAT IT'S FOR	READ OR ACT?	WHO	LIFECYCLE
Building & shipping outbound orders.	Full action screen.	DISPATCH ADMIN	Draft → Packing → Shipped.

The shipment lifecycle

STATUS	MEANING
Draft	Created, manifest being set. Can be edited or deleted.
Packing	Boxes open, units being packed. Can be edited or cancelled.
Ready	Fully packed against the manifest, not yet shipped (a green ✓ Ready badge).
Shipped ✓	Sent out: all its units marked shipped.
Cancelled ✕	Called off: its units return to the dispatch pool.

Building a shipment, step by step

- 1
- + **New Shipment.** Give it an optional title, pick a **channel** (required), set a scheduled date, and build the **manifest**: add each product and enter the quantity per model/colour variant. A live total shows at the bottom. Create it.
- 2
- Open the shipment** (click its row). You'll see meta cards (Channel, Pool/Target, Packed, Scheduled), the manifest with progress bars, and the boxes list.
- 3
- + **Add Boxes.** Open one or more packing boxes. Units get scanned into boxes on the floor; each box shows its unit count and status.
- 4
- Manage boxes as needed.** Expand a box to see its units; ✕ **Remove** a unit (it returns to the pool), **Reprint** a label, **Reopen** a packed box, or ✕ **Remove** an empty box.

- 5 **Shipped.** When everything's packed, mark the shipment shipped; this updates every unit in it to Shipped.

Reading the numbers

Pool / Target	Units allocated to this shipment so far versus the manifest goal (e.g. 8 / 20). This is allocation progress, not the same as packed.
Packed	Units actually packed into boxes: green when 100% of a manifest line is packed, yellow while in progress.
Box status	Open (amber) → being packed; Packed / Closed (green ✓) → done.

⚠ EDITING LIMITS

You can only edit a shipment while it's Draft or Packing, and you can't set a manifest line below what's already packed, nor remove a product that has packed units. Deleting is for drafts only.

🚚 SHIPPING IS THE COMMITMENT

Mark Shipped flips every unit in the shipment to Shipped; do it only when the consignment has genuinely left. Cancelling returns units to the dispatch pool (not lost), but "this cannot be undone", so cancel deliberately.

Delivery Challans

/dispatch-challans

DISPATCH

ADMIN

Delivery Challans are the transport documents for outbound goods movements: GST-style goods-movement notes numbered `LOT-DC-YYYY-NNNN`. This screen is the register where you find, create, issue and print them.

WHAT IT'S FOR	READ OR ACT?	WHO	LIFECYCLE
Creating & managing delivery challans.	Action: create / issue / cancel / print.	DISPATCH ADMIN	Draft → Issued (or Cancelled).

The register

The list shows Challan No., Date, Dispatched To, Purpose, Qty, Total (₹), **EWB** and Status. Filter by status (with live counts), search by number or recipient, and set a date range. Status badges: Draft (neutral), Issued (green ✓), Cancelled (red ✕). The EWB column reads `—` if no e-way bill is needed, a blue `✓ {number}` when recorded, or an amber `⚠ REQUIRED` when needed but missing.

Creating a challan

- + New Challan.** Set the date, purpose (Material transfer, Job work, Sample, Return for repair...) and GST rate.
- From & To.** Pick the dispatching company address (prefilled, editable) and enter the recipient's name, address and GSTIN.
- Goods supplied.** Add a line per item (search a product to auto-fill its description and HSN, or type freehand) with quantity, unit and rate. Amounts total automatically.
- Transport & e-way bill (optional).** Add mode, vehicle and transporter, and EWB number/date if you have one.
- Save as Draft or Issue Challan.** Saving as draft lets you review later; issuing locks it from edits. The number is assigned on save.

On a challan

From a challan's detail page you can **Print** it (an A4 document that opens and prints in a new tab), **Issue** a draft (locks it), or **Cancel** it with a reason.

① THE ₹50,000 E-WAY BILL LINE

When a challan's total reaches **₹50,000**, an amber banner reminds you an e-way bill is required (a quieter note appears from ₹40,000). The EWB is generated separately at the government portal; the banner is a reminder. You can still save and issue without entering one, but the register will keep flagging it **⚠ REQUIRED** until you do.

⚠ ISSUING LOCKS IT

Once issued, a challan can't be edited, only printed or cancelled. Get the details right while it's still a draft.

Channel Master

/dispatch-channels

ADMIN

Channel Master is the list of every dispatch/sales channel (Amazon, Flipkart, Cred, BOM Bulk, retail partners and so on) that shipments and units are assigned to. It's the small but important screen that defines the choices everywhere else in Dispatch.

WHAT IT'S FOR	READ OR ACT?	WHO	NO DELETE
Maintaining the channel list.	Action: add / edit channels.	ADMIN	Deactivate instead.

A channel's settings

Name	The channel's name (e.g. Amazon, Flipkart, BOM Bulk).
Type	Ecom, Retail or Other, shown as a colour badge across Dispatch.
Fulfillment	Unit (dispatched per unit) or Bulk (warehouse transfer). Bulk channels make a "Destination Warehouse" field appear when you create a shipment for them.
Sale channel	Whether the channel generates revenue. This drives the green "sold" colouring and totals on the Dispatch Overview.
Active	Whether the channel is offered when creating shipments. Inactive channels show dimmed.

Adding & editing

+ **Add Channel** opens an inline form; new channels are always created active. **Edit** on any row lets you change its type, fulfillment, sale flag and active state.

① RETIRE, DON'T DELETE

There's no delete. A channel that's no longer used is switched to **Inactive** so its history stays intact and it simply stops appearing as a choice for new shipments.

Repack Runs

[/repack-runs](#) DISPATCH SUPERVISOR ADMIN

Repack Runs track “channel swaps”: when already-packed units need re-doing for a different sales channel (say retail → ecom). A run sets a target number of units to repack; operators then physically scan boxes at the Repack In and Repack Out stations, and progress accrues against the run. This screen opens, tracks and closes those runs; the actual repacking happens on the scanner.

WHAT IT'S FOR	READ OR ACT?	WHO	REPACKING
Tracking channel-swap repacks.	Create / complete / cancel runs.	DISPATCH SUPERVISOR ADMIN	Done at the scanner stations.

How a channel swap works

Request the run from **New Run / Request → Repack** (product, from → to channel, quantity). That raises two requests automatically: the store issues the **to-channel packaging** (box and tray) into the Issue Queue, and dispatch gets a **release list**. Then the physical work happens on the floor:

- 1 Dispatch releases the units at Repack Release.** A dispatch operator scans each box to hand it over to the run. Units still on the production side (just out of PKG Out) skip this.
- 2 Scan the old box at Repack In.** Only released (or PKG-Out) units are accepted; the unit rolls back to QC Pass and everything downstream is wiped, ready to re-box for the new channel.
- 3 Re-pair at Repack Out.** The operator re-confirms the car + remote pair; the new channel comes from the request and a fresh sticker prints.
- 4 The unit rejoins dispatch.** From there it flows through the normal pipeline (PKG Out → allocate → pack → dispatch) unchanged. The old packaging goes back to the store via line flush.

Tracking it from this screen

This screen tracks the runs requested from New Run / Request. The list shows each run's progress (`repacked / target + %`) and status: **Open**, **In Progress**, **Completed** (green), **Cancelled**. A run's detail page tracks each swap as it happens: car UPC, paired remote, the `from → to` channel, old and new labels, and in/out times. Units scanned in but not yet re-packed out show as **in-flight**. **Complete** closes the run; **Cancel** stops it (units already repacked stay repacked).

① REQUESTS COME FROM NEW RUN / REQUEST

Repack runs are now requested with a product, channel direction and quantity (not just a bare count), which is what lets the store auto-receive the right packaging and dispatch know which units to release. This screen is where you watch and close them.

PART 6

Repair

Tracking customer repairs and the floor repair queue.

Customer Repairs

/customer-repairs

SUPERVISOR

ADMIN

Customer Repairs is the tracker for warranty and repair cases coming back from customers. Each case (numbered `CR-001`) moves through seven clear stages from “request received” to “dispatched”, so anyone can see where a customer’s unit is and how long it’s been there.

WHAT IT’S FOR	READ OR ACT?	WHO	STAGES
Tracking customer repair cases.	Action: capture, edit, advance.	SUPERVISOR ADMIN	Seven, in order.

The seven stages

1. **Request received:** case logged.
2. **Pickup requested:** courier pickup arranged.
3. **Pickup done:** collected from the customer.
4. **Reached stores:** arrived at the warehouse.
5. **Handed to production:** with the floor for repair.
6. **Repaired · Ready:** fixed, ready to send back.
7. **Dispatched:** sent back to the customer (closed).

The list

Every case with its CR number, customer (name + phone), Order ID, AWB, channel, current stage and **Age**. Age is colour-coded so stale cases stand out: under 3 days grey, 3–6 days amber, 7+ days red. Filter by stage (each chip shows a count) and search by customer, order, AWB or CR number.

Capturing and working a case

- 1 **+ New Repair.** Only the **customer name** is required; capture fast, fill logistics (order, AWB, pickup) later. New cases start at “Request received”.
- 2 **Open the case** to edit its details (Save Changes appears once you’ve edited something) and read its event history.
- 3 **Advance to next stage** with the big yellow button, optionally adding a note. Or use the dropdown to **jump** ahead several stages at once.

① **EVERYTHING IS LOGGED**

The case's history records who created it, every stage change (with any note) and which fields were edited, so the full story of a customer's repair is always there. The three repair tiles on the **Returns** screen link straight into this list filtered by stage.

Repair Queue

/repair-queue

SUPERVISOR

ADMIN

The Repair Queue is the pool of physical units on the floor waiting for a repair run: units that failed QC or came back as returns and need rework. Open it to see how many units of each product are waiting so a repair run can be planned.

WHAT IT'S FOR	READ OR ACT?	WHO	RUNS CREATED IN
Seeing units awaiting repair.	Read-only.	SUPERVISORADMIN	The Store (Garage) system.

Reading it

A table grouped by product (each product shows a subtotal), then variant rows with Model, Colour, Status and Count, plus a running total at the top. The status tells you why a unit is in the pool:

STATUS	MEANING
QC Fail	Failed the quality check and was pulled for rework.
RTO In	A return that came back in and needs repair.
Scrapped	Marked scrapped during repair.

① THIS IS A VIEW, NOT A WORKBENCH

The Repair Queue shows what's waiting; the actual repair runs that consume the pool are created in the Store (Garage) system. Don't confuse this with **Customer Repairs**: that tracks customer warranty cases, while this is the floor's pool of physical units to rework.

PART 7

Reporting

Cross-cutting production and dispatch reports.

Reporting

/reporting

ADMIN

Reporting is the analytics hub. Where the Dashboard and Lines show *today*, Reporting lets you look across a date range (production versus target, cycle times, defect and quality trends, line throughput) and export the data as CSV for audits and deeper analysis.

WHAT IT'S FOR	READ OR ACT?	WHO	RANGE
Period reports & data exports.	Read; downloads CSVs.	ADMIN / managers.	10 Days · Week · Month · Custom.

The five sections

SECTION	WHAT IT SHOWS
Production	QC pass, first-pass yield, dispatched, vs-target and run counts, with a daily bar chart and a By Product / By Line table.
Cycle Time	Average QC, PKG and RTD cycle times, with median, fastest/slowest and outlier counts. Green up to 30 min, yellow to 60, red beyond.
Defects	Total occurrences, FPY, top defect codes (coloured by severity) and a severity split, By Code or By Product.
Throughput	Per-station takt times and a per-line bottleneck view that flags the slowest station on each line.
Downloads	CSV exports (see below).

Set the range once at the top (10 Days, This Week, This Month, or Custom) and it applies across the sections.

Downloads

Grouped CSV exports: **Production** (QC View, Plan vs Actual, Defects), **Audit & Compliance** (Customer Repairs, Process Deviations, QC Audit Findings, Scan Violations, Unit Restocks, Damage/Scrap Ledger), and **Movement & Issuance** (Direct Issuances, Cycle Counts, Stock Adjustments). Each downloads for the selected range; an empty range tells you there were no rows.

① OPENING THE CSVS

Files are UTF-8 with a BOM, so Excel opens them cleanly. For richer per-module views, the footer points you to **Garage → Reports**. One export (**Stock Adjustments**) needs the finance reporting permission and is otherwise disabled.

PART 8

Admin

UPC generation, operator management and printing.

UPC Generator

/upc ADMIN

The UPC Generator mints and prints sheets of barcode stickers for new units: the LOT-... codes that follow each car and remote through its whole life. Open it when the floor needs fresh stickers, and use it to track each print batch from generated to received.

WHAT IT'S FOR	READ OR ACT?	WHO	SHEET
Generating & printing UPC stickers.	Action: generate, print, track.	ADMIN	A3, 609 stickers per sheet.

Generating a batch

- 1 Pick the product.** Search by code or name. Products with a remote show a + REMOTE tag; if so, choose the **Car** or **Remote** component (Remote appends an "R" to the code).
- 2 Set the quantity** (1–10,000) and any notes.
- 3 ⚡ Generate & Print.** The batch is created (you'll see its ID, sticker count, the sequence range (CODE-from → CODE-to) and the number of A3 sheets), and the print window opens automatically a moment later.

Tracking batches

Recent Batches lists the last 50, each with its status. The lifecycle:

STATUS	MEANING & BUTTON
Generated	Created. Mark Sent when it goes to the printer.
Sent to Print	At the printer. Mark Received when the printed sheets are back.
Printed	Printed; mark Received once collected.
Received	Sheets in hand: done.

 **Print** on any row reprints that batch's sheet (an A3 grid of 13 mm QR stickers, each encoding the unit's LOT code).

 NUMBERS ARE MINTED ONCE

Each batch draws a fresh, never-reused block of LOT numbers from the shared sequence: the single source of truth for unit numbering. Generate only what you'll actually print and apply; a generated batch consumes those numbers whether or not the stickers are used.

Operators

/operators

ADMIN

Operators is the floor staff roster: add and edit people, deactivate leavers, and print each operator’s attendance QR identity card. It’s the master list that attendance, rostering and performance all draw from.

WHAT IT’S FOR	READ OR ACT?	WHO	DEPARTMENTS
Managing operators & their QR cards.	Managers edit; anyone can print QR.	ADMIN (+ all can view/print)	Assembly, QC, Packaging, Admin, Store, Dispatch.

The roster

A table of Name, Employee ID, Department, Type (In House / Contract) and Status. Search by name or ID, filter by department and status (defaults to Active). Inactive operators show dimmed.

Managing people

- **+ Add Operator:** name is required; pick a department and employment type. The Employee ID is assigned by the system once the operator exists.
- **Edit:** update department, status and phone.
- **Deactivate:** for someone who’s left; confirm the prompt. They drop out of the active list and pickers but their history is kept.
-  **QR:** available to everyone; prints a name-badge card with the operator’s attendance QR.

⚠ TWO DIFFERENT QR CARDS

The QR printed here encodes the **employee ID** and is for the **attendance** scanner (clocking in/out). It is *not* the scanner-station login card used to log into a workstation; that’s a separate card. Don’t hand someone an attendance card expecting it to log them into a station.

Print Center

/print

ADMIN

The Print Center is for reprinting a unit’s shipping/packing label when the original is lost, damaged or jammed. Look up an already-packed unit by its batch label or car UPC, pick which printer should produce it, and re-queue the label.

WHAT IT’S FOR	READ OR ACT?	WHO	PRINTERS
Reprinting packing/shipping labels.	Action: looks up & re-queues prints.	ADMIN	L1, L2, L3, D1, D2.

Reprinting a label

- 1

Choose Single or Bulk. Single looks up one unit; Bulk takes a list (one label/UPC per line).
- 2

Search. Enter a batch label or car UPC: a bare number is auto-padded to a full `L0T-...`; a value ending in `-E` or `-R` is treated as a batch label.
- 3

Pick the printer. Select a destination (L1–L3, D1, D2) under “Print To”. You must choose one before printing.
- 4

 **PRINT** on a found row (or **PRINT ALL**) re-queues the label to that printer.

▲ “QUEUED” MEANS SENT, NOT PRINTED

A green `Queued` badge confirms the job was sent to the printer’s queue; it doesn’t guarantee paper came out. Check the physical printer. Entries that can’t be matched collect in a red “Not found” box; PRINT ALL skips rows already queued so you can’t double-print by re-clicking.